

Yuxiang Wei

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Education

PhD	Georgia Institute of Technology, <i>Electrical and Computer Engineering</i> , advised by Vince Calhoun	2028.5
MS	Georgia Institute of Technology, <i>Electrical and Computer Engineering</i> , advised by Vince Calhoun	2025.5
Research Focus: machine learning, multimodal brain modeling, Brain-AI alignment		
BS	University of Electronic Science and Technology of China, <i>Electronic Information Engineering</i>	2023.6

Technical Skills

- Programming and Tools:** Python, Pytorch, Sci-kit Learn, C++, MATLAB, CUDA, Docker, Git, Linux, AWS.
- Machine Learning:** SVM, PCA, ICA, kernel methods, L1/L2 regularization, k-means.
- Deep Learning:** GNN, Transformer, SSM, GAN, VAE, diffusion models, representation learning.

Experience

Amazon – Applied Scientist Intern Austin, TX | 2025.5-2025.8

Graph-based Time-Series Probabilistic Forecasting

- Conducted in-depth analysis on the over-smoothing issue of current GNN models and proposed alternative Transformer-based model with graph-derived positional encoding to exploit **graph-based inductive bias**.
- Architected a novel **spatiotemporal diffusion model** to generate full probabilistic forecasts across a 42,000+ node graph, moving beyond deterministic prediction to quantify uncertainty and capture complex futures.
- Innovated a **counter-factual** forecasting framework to generate robust predictions for out-of-distribution data; optimized large-scale training and inference, outperforming production models by over **30%** on WAPE.

TReNDS Center, Georgia Institute of Technology – Graduate Research Assistant Atlanta, GA | 2023.8-present

Neural-Inspired Framework for Brain Visual Decoding

- Developed a novel hierarchical **Mixture-of-Experts (MoE)** architecture to decode fMRI signals, mirroring the brain's functional specialization and improving model interpretability for neuroscientific insight.
- Engineered a dynamic space-time routing module to guide a **diffusion model**, achieving image generation from brain activity while enabling efficient **cross-subject generalization** with about **90%** fewer tunable parameters.

Foundational fMRI-Language Model

- Pioneered **multimodal large language model (MLLM)** architectures for unified fMRI understanding; leveraging LLM's language ability to **reason** over complex brain activation patterns.
- Designed an end-to-end pipeline to **fine-tune** pretrained LLMs on fMRI tokens via modality projection and next-token prediction; benchmarked on multiple downstream tasks and achieved SOTA.

Tencent – Applied Scientist Intern

Shenzhen, China | 2024.5-2024.8

Multi-treatment Multi-task Uplift Modeling

- Explored **causal inference** and **uplift modeling** to boost user activity; developed a novel uplift modeling method that enables direct uplift prediction across multiple treatments and tasks.
- Identified biases in existing methods due to non-random data collection; designed a new uplift estimation method to reduce bias, resulting in a more than **20%** increase in monthly active users.

Other Projects

Unified Framework for Agentic Memory System

Quanta Alpha

- Designed a **unified framework long-term memory system** that decomposes memory systems into modular components (representation, organization, retrieval); Identified critical insights through controlled experiments, revealing that **architectural complexity** often yields diminishing returns compared to optimized foundational settings in **memory maintenance** and **retrieval** pipelines.

Benchmark for AI Clones

Quanta Alpha

- Introduced **CloneMem**, a novel benchmark for **AI Clones** simulating **long-term memory** through **non-conversational digital traces**; Conducted extensive benchmarking of **memory mechanisms** on tasks requiring the retrieval of evolving **experiences and opinions**, demonstrating that existing models struggle with the **temporal coherence** required for realistic AI clone interaction.

Publications

Google Scholar: scholar.google.com/citations?user=iGde1EAAAAAJ

[1] Yuxiang Wei, Yanteng Zhang, Xi Xiao, Chengxuan Qian, Tianyang Wang, Vince D. Calhoun “fMRI-LM: Towards a Universal Foundation Model for Language-Aligned fMRI Understanding”, Accepted to *CVPR 2026*.

- [2] Connor Lane, Mihir Tripathy, ... **Yuxiang Wei**, ..., Paul S. Scotti, "Scaling Vision Transformers for Functional MRI with Flat Maps", Accepted to *ICML 2026*.
- [3] **Yuxiang Wei**, Yanteng Zhang, Xi Xiao, T Wang, Vince D, Calhoun, "MoRE-Brain: Routed Mixture of Experts for Interpretable and Generalizable Cross-Subject fMRI Visual Decoding". *Advances in Neural Information Processing Systems (NeurIPS)*, 2026, 38: 55870-55902.
- [4] Sen Hu[†], **Yuxiang Wei**[†], JiaXin Ran, et al. "Does Memory Need Graphs? A Unified Framework and Empirical Analysis for Long-Term Dialog Memory", Accepted to *ACL 2026 Main Conference*.
- [5] Sen Hu[†], Zhiyu Zhang[†], **Yuxiang Wei**[†], Xueran Han, et al. "CloneMem: Benchmarking Long-Term Memory for AI Clones", Accepted to *ACL 2026 Main Conference*.
- [6] **Yuxiang Wei**, Yanteng Zhang, Anees Abrol, Vince D. Calhoun, "4D Multimodal Co-attention Fusion Network with Latent Contrastive Alignment for Alzheimer's Diagnosis." *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. 2026: 5457-5466.
- [7] Xi Xiao, Yunbei Zhang, Janet Wang, Li Zhao, **Yuxiang Wei**, Tianyang Wang, "RoadBench: A Vision-Language Foundation Model and Benchmark for Road Damage Understanding." *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. 2026: 6016-6026.
- [8] Xi Xiao, Yunbei Zhang, Xingjian Li, Tianyang Wang, Xiao Wang, **Yuxiang Wei**, Jihun Hamm, Min Xu, "Visual Instance-aware Prompt Tuning", *Proceedings of the 33rd ACM International Conference on Multimedia (ACM MM)*. 2025: 2880-2889.
- [9] **Yuxiang Wei**, Anees Abrol, Deqiang Qiu, James Lah, Allan I Levey, Vince D Calhoun, "Spatiotemporal State Space Modeling of Dynamic Brain Connectivity in Cognitively Normal Individuals at Risk for Alzheimer's Disease", *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2026: 1516-1520.
- [10] **Yuxiang Wei**, Annes Abrol, Vince D. Calhoun, "From Symptomatic to Pre-symptomatic: Adaptive Knowledge Distillation for Early Alzheimer's Detection." *IEEE Transactions on Biomedical Imaging*. 2025.
- [11] **Yuxiang Wei**, Annes Abrol, Vince D. Calhoun, "Hierarchical Spatio-Temporal State-Space Modeling for fMRI Analysis." *International Conference on Research in Computational Molecular Biology. Springer (RECOMB)*, 2025: 86-98.
- [12] **Yuxiang Wei**, Zhaoxin Qiu, Yingjie Li, Yuke Sun, Xiaoling Li, "Multi-Treatment Multi-Task Uplift Modeling for Enhancing User Growth." arXiv preprint arXiv:2408.12803, 2024.
- [13] **Yuxiang Wei**, and Huan Wang. "Wavelet Integrated Attention Network with Multi-Resolution Frequency Learning for Mixed-Type Wafer Defect Recognition." *Engineering Applications of Artificial Intelligence*, vol. 121, May 2023.
- [14] **Yuxiang Wei**, Yuqian Chen, Tengfei Xue, Fan Zhang, and Lauren J. O'Donnell. "An Explainable Multi-View Deep Network for Prediction of Non-Imaging Phenotypes using Anatomical Multi-view Data." *The Medical Image Computing and Computer Assisted Intervention Society (MICCAI) workshop* 2023.
- [15] **Yuxiang Wei**, and Huan Wang. "Mixed-Type Wafer Defect Recognition with Multi-Scale Information Fusion Transformer." *IEEE Transactions on Semiconductor Manufacturing*, vol. 35, no. 2, pp. 341-352, May 2022.
- [16] **Yuxiang Wei**, and Huan Wang. "Mixed-type Wafer Defect Pattern Recognition Framework based on Multi-faceted Dynamic Convolution." *IEEE Transactions on Instrumentation and Measurement*, vol. 71, pp. 1-11, 2022.

[†] denotes equal contribution